Project Title: Doctor Appointment Booking & Patient File Management System

Document Version: 1.0.0

Classification: Technical Architecture & Specifications

1. System Overview
The Doctor Appointment Booking & Patient File Management System is an enterprise-grade, cloud-ready healthcare application designed to optimize clinical workflows, manage Electronic Health Records (EHR), and automate patient-doctor scheduling. The platform operates on a Three-Tier Architecture model to enforce clean separation of concerns, high scalability, and robust security protocols required for handling sensitive medical data.

2. Architectural Blueprint
The system is split into three decoupled operational layers:

2.1. Presentation Layer (Frontend)
Technology Stack: React.js, JavaScript (ES6+), HTML5, CSS3.

Core Responsibilities: * Renders responsive user interfaces for Patients, Doctors, and Administrators.

Manages client-side state using state hooks to accurately display real-time availability without explicit page reloads.

Handles asynchronous HTTP/HTTPS processing to communicate with backend APIs.

2.2. Application Layer (Backend Engine)
Technology Stack: Django (Python) / Node.js framework.

Core Responsibilities:

Exposes secure RESTful API endpoints for data exchange.

Executes core business logic (e.g., verifying booking slot availability, calculating schedule conflicts).

Implements secure, token-based authentication (JSON Web Tokens / JWT) and manages user sessions.

2.3. Data Storage Layer (Database)
Technology Stack: Relational Database Management System (PostgreSQL / MySQL / SQLite).

Core Responsibilities:

Maintains ACID-compliant structured data tables.

Preserves relational integrity between entities (Patients, Doctors, Appointments, MedicalRecords) using Primary Key (PK) and Foreign Key (FK) constraints.